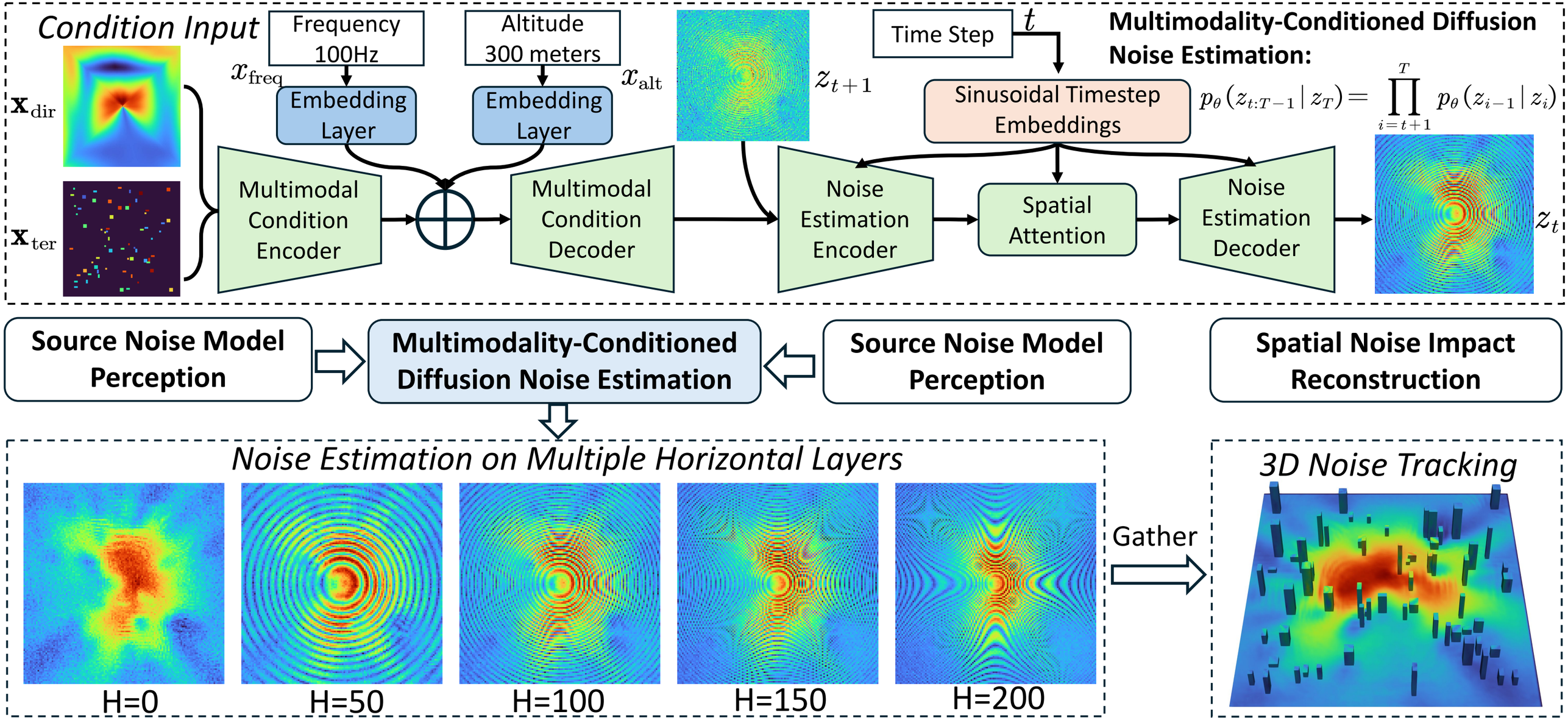
Condition Input
Frequency 100Hz
Altitude 300 meters
x_{freq}
x_{alt}
$\mathbf{x}_{dir}$
$\mathbf{x}_{ter}$
Embedding Layer
Embedding Layer
Multimodal Condition Encoder
Multimodal Condition Decoder
z_{t+1}
Time Step
t
Multimodality-Conditioned Diffusion Noise Estimation:
$p_\theta(z_{t:T-1} \mid z_T) = \prod_{i=t+1}^{T} p_\theta(z_{i-1} \mid z_i)$
Sinusoidal Timestep Embeddings
Noise Estimation Encoder
Spatial Attention
Noise Estimation Decoder
z_t
Source Noise Model Perception
Multimodality-Conditioned Diffusion Noise Estimation
Source Noise Model Perception
Spatial Noise Impact Reconstruction
Noise Estimation on Multiple Horizontal Layers
H=0
H=50
H=100
H=150
H=200
Gather
3D Noise Tracking